

C# Fundamentals

Getting Started with C# and .NET

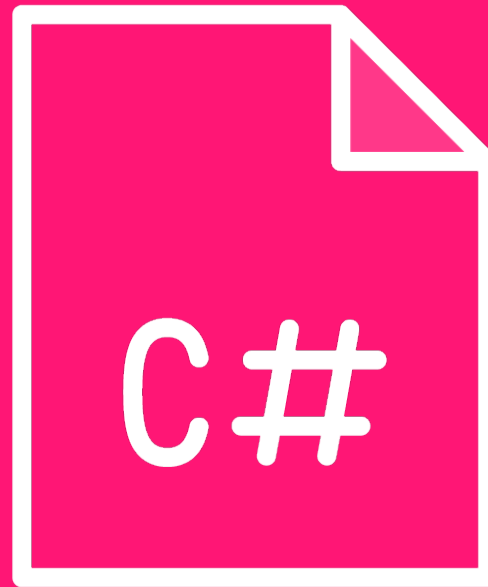


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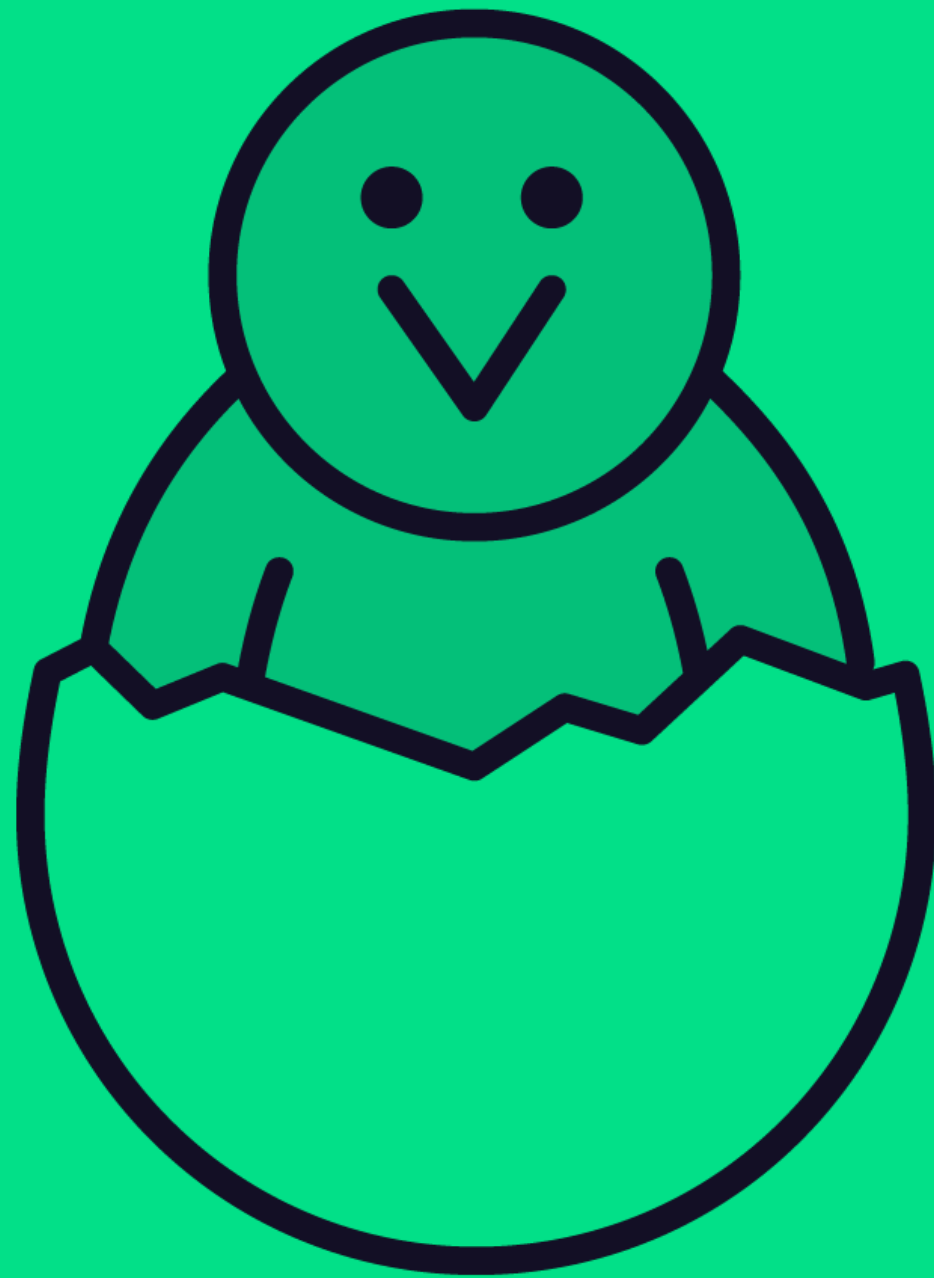




Learning C# is a great choice!

C# is part of .NET





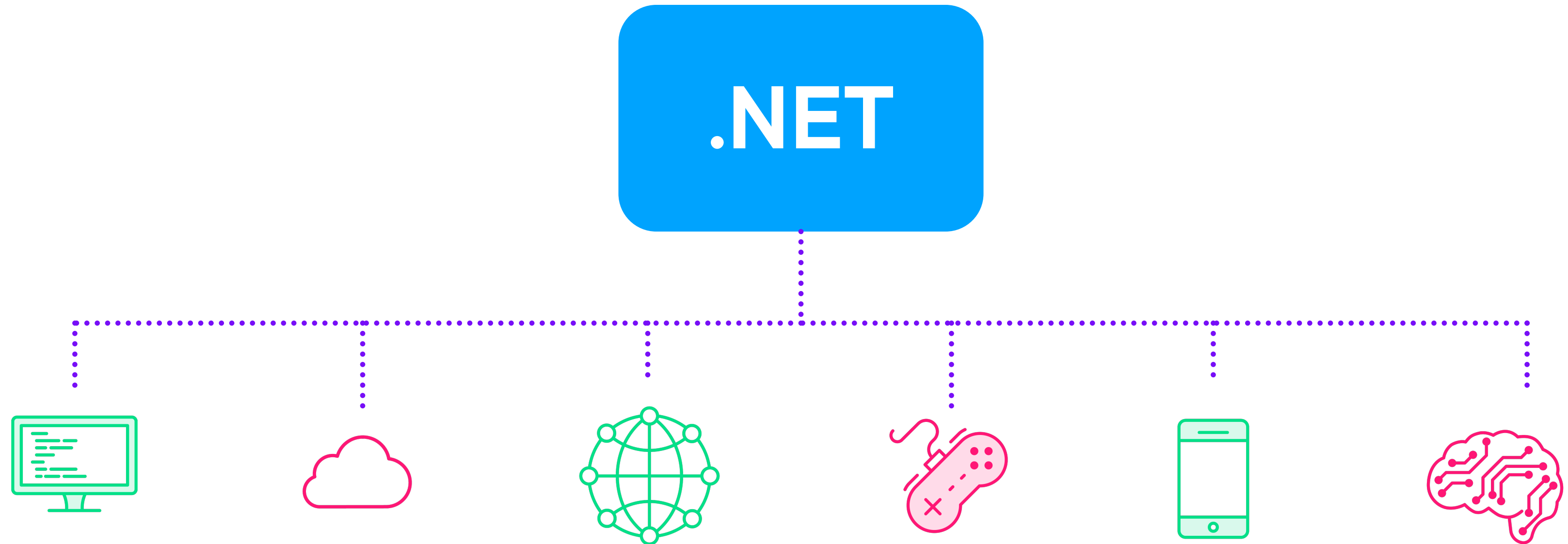
Hello .NET

Developer platform

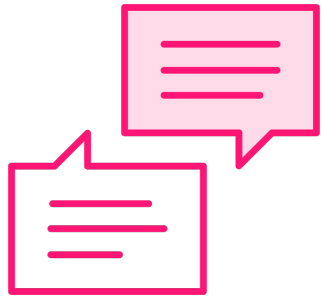
Open source



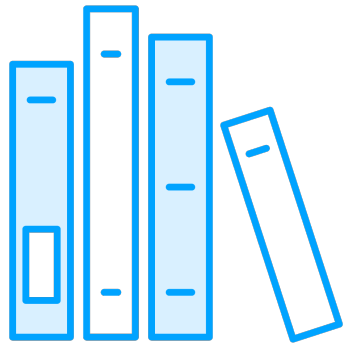
The Unification of .NET



The .NET Ecosystem



Languages



Libraries



Tools





.NET 8

Cross-platform





The Languages of .NET

C#

Visual Basic

F#



A

B

C

**No C# knowledge
is required!**



What Is Programming?

★★★★★ By Simon Allardice

This is an accessible, language-agnostic course on the fundamentals of programming. Learn how to get started programming - in any language.



Bethany's Pie Shop HR

- HR Management software





**Follow along
with everything I'm
doing!**



Downloads for This Course

Table of contents	Description	Transcript	Exercise files	Discussion	Related Courses
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Version Check



This version was created by using:

- C# 12
- .NET 8
- Visual Studio 2022 Community Edition



Version Check



This course is 100% applicable to:

- C# 10, C# 11 and C# 12
- .NET 6, .NET 7 and .NET 8
- Visual Studio 2022 (any edition)

Most code will work fine with older versions of C#



Overview



Hello C#

Setting up your environment

Building your first C# program

Debugging our code

Building applications using VS Code and the CLI

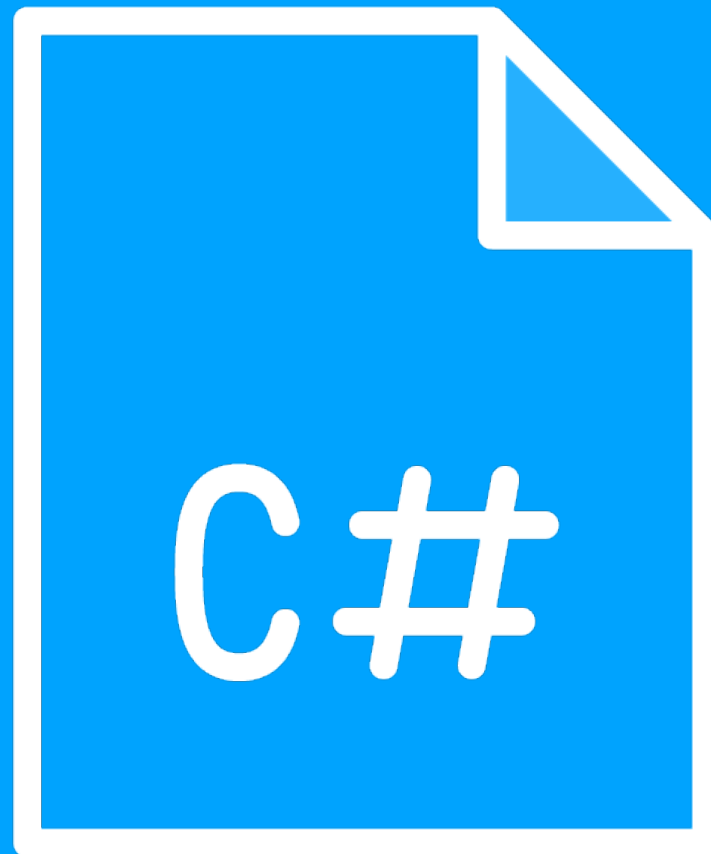
Browsing the docs





Hello C#





Hello C#

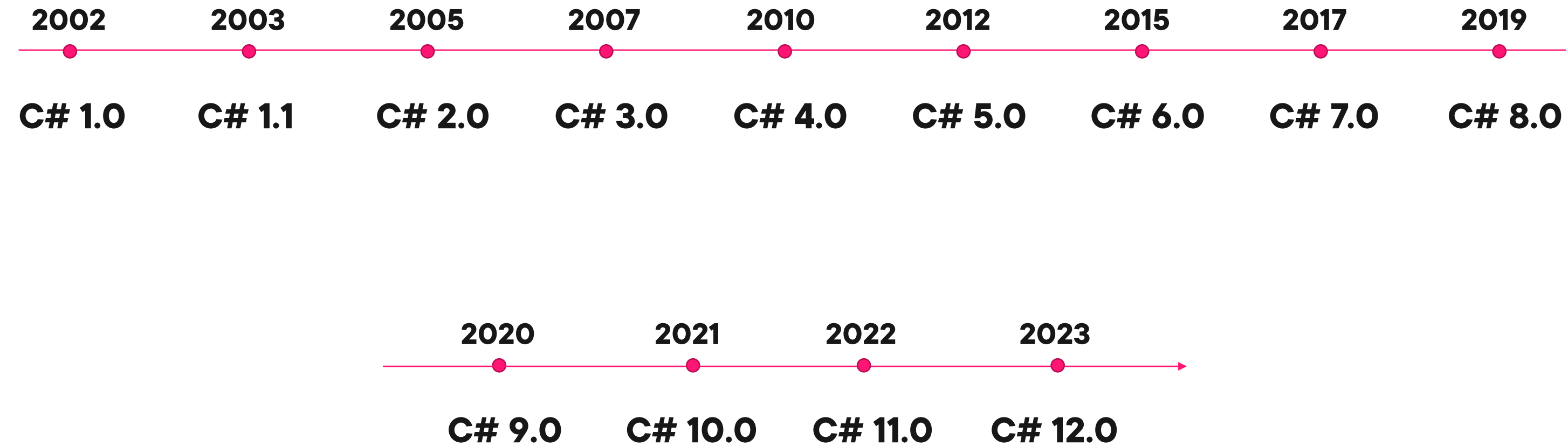
Main language for .NET development

Object-oriented and type-safe
programming language

Familiar for C, C++, Java and JS
developers



C# Is Actively Maintained



Hello C#

Console applications

Desktop applications

Web applications

Mobile applications

Services



Setting up Your Environment



Building .NET Applications with C#

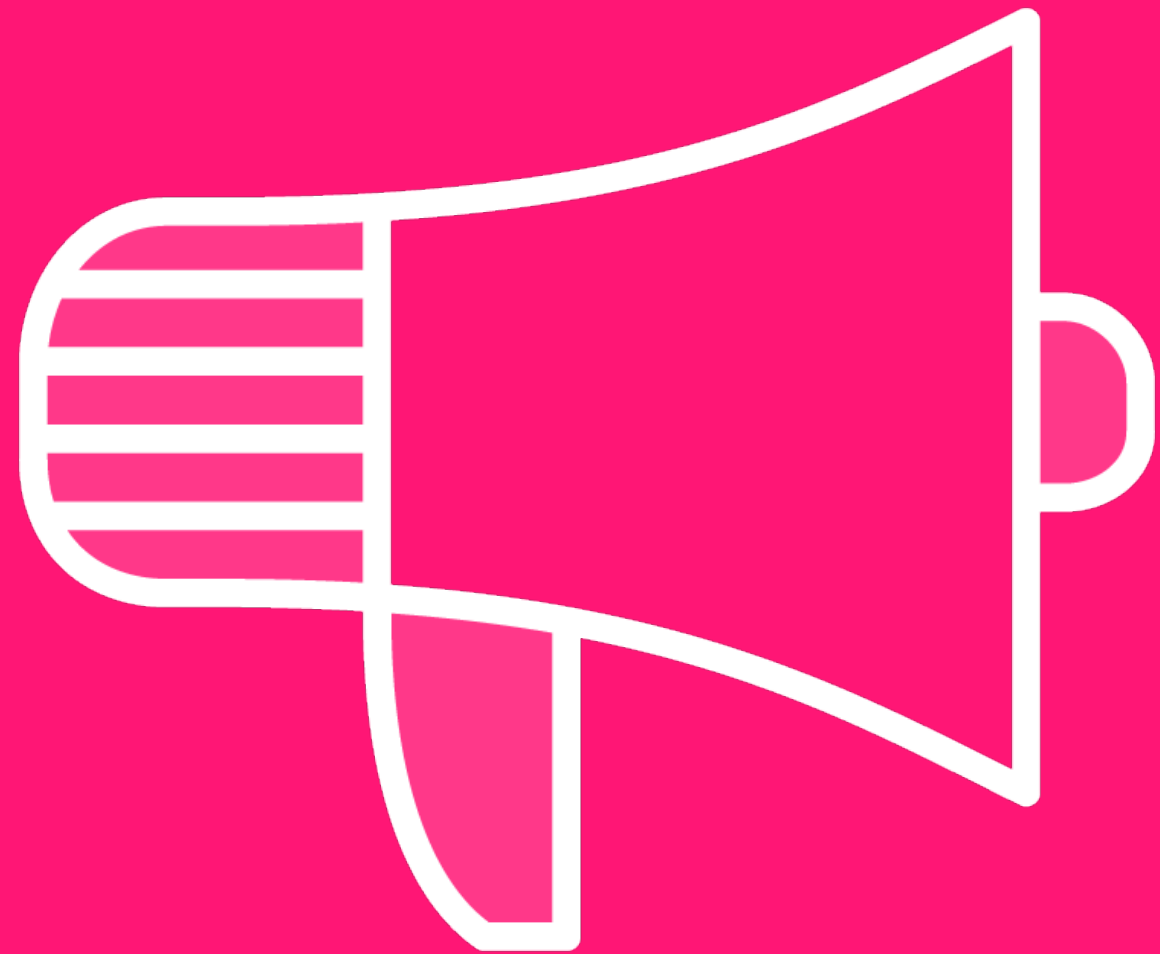
Visual Studio 2022

Windows

**.NET CLI &
Visual Studio Code**

All platforms





Introducing Visual Studio 2022

Flagship IDE
(Integrated Development Environment)

Many features



Visual Studio 2022 Editions

Community

Free to use

Professional

Paid

Aimed at smaller
teams

Enterprise

Paid

Aimed at larger
teams



Demo



**Setting up your environment
for C# development**

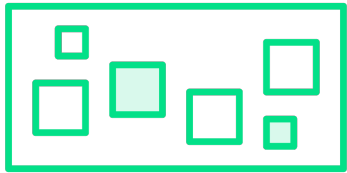




Building Your First C# Program Using Visual Studio



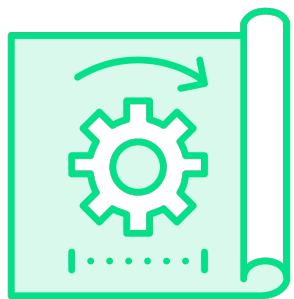
Using Projects



Containers for code files



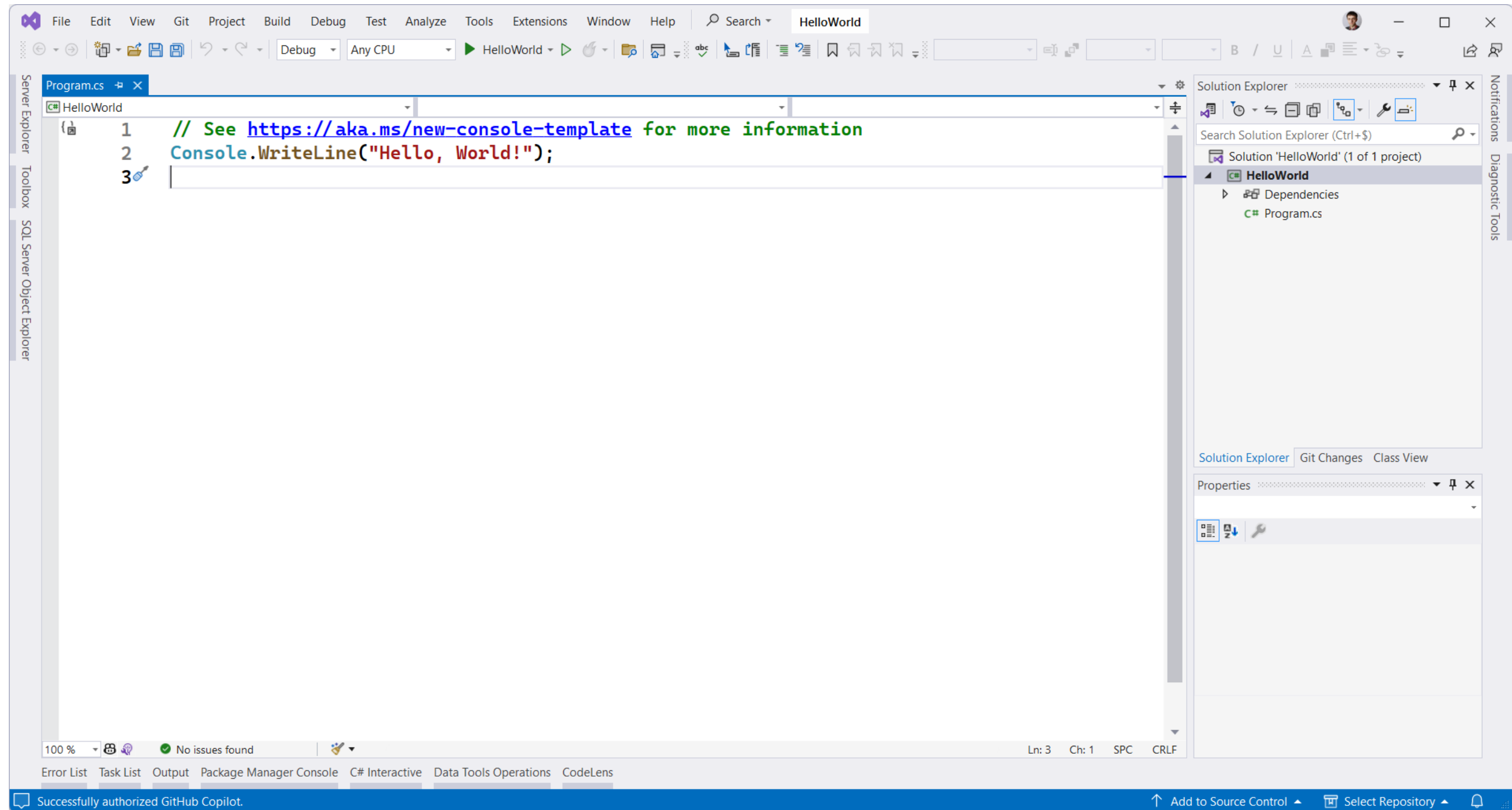
Compiled into executable files (assembly)



Different templates



The Visual Studio 2022 Interface



Demo



Creating your first C# application

Running the application



Demo



Writing C# code

Fixing a syntax error



Demo



Looking at the generated files

- Project
- Solution
- Executable file

Executing our compiled application





Debugging Our Code



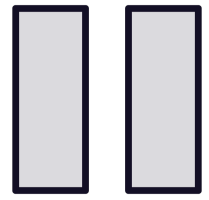


We will make errors!

Debugging using Visual Studio



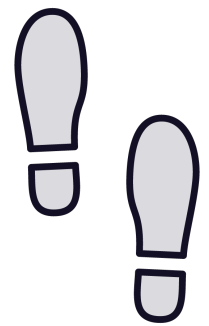
Introducing Breakpoints



Pause the running code



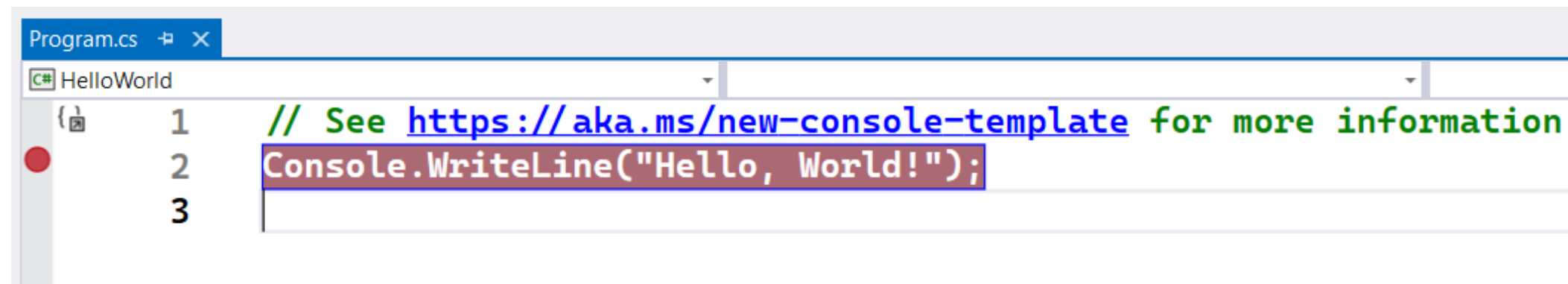
Inspect the state of the running application



Step through the different lines of code



Adding a Breakpoint



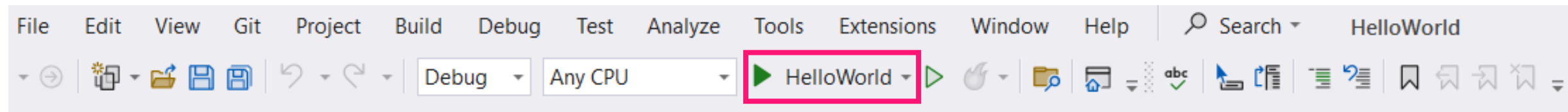
The screenshot shows a code editor window with a tab labeled "Program.cs". The code is in C# and is part of a "HelloWorld" project. The code consists of three lines:

```
1 // See https://aka.ms/new-console-template for more information
2 Console.WriteLine("Hello, World!");
3
```

A red circular breakpoint is placed on the left margin, aligned with line 2. The text on line 2, `Console.WriteLine("Hello, World!");`, is highlighted with a blue selection background.



Running with the Debugging Attached



Demo



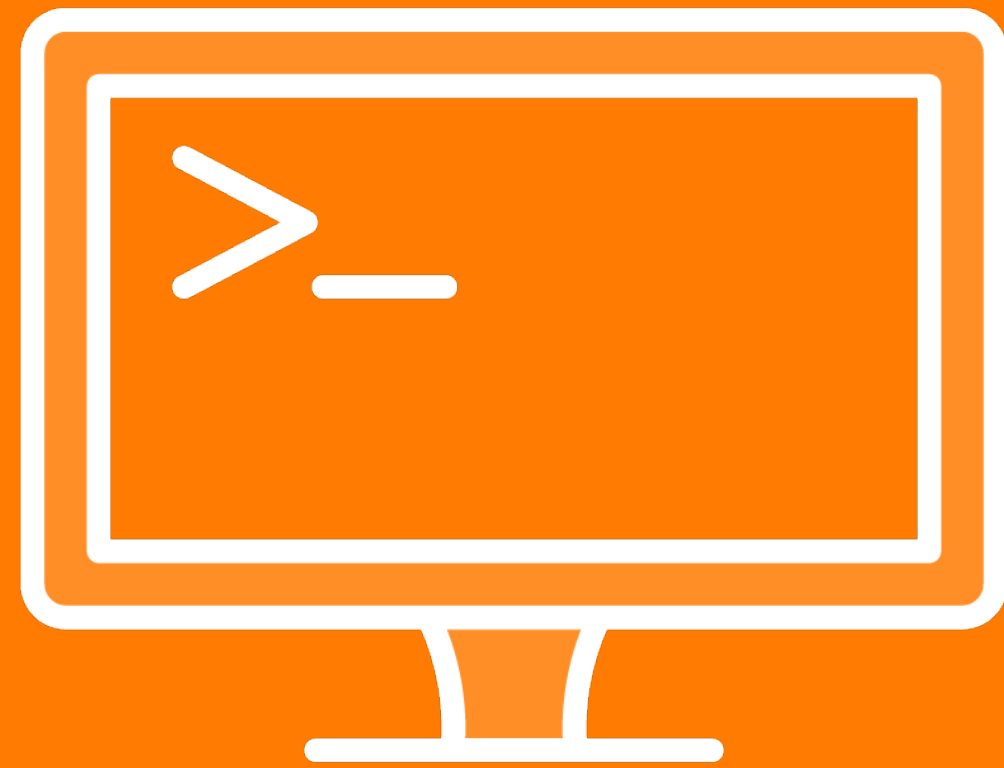
Understanding the debugger





Building applications using VS Code and the CLI





Understanding the CLI

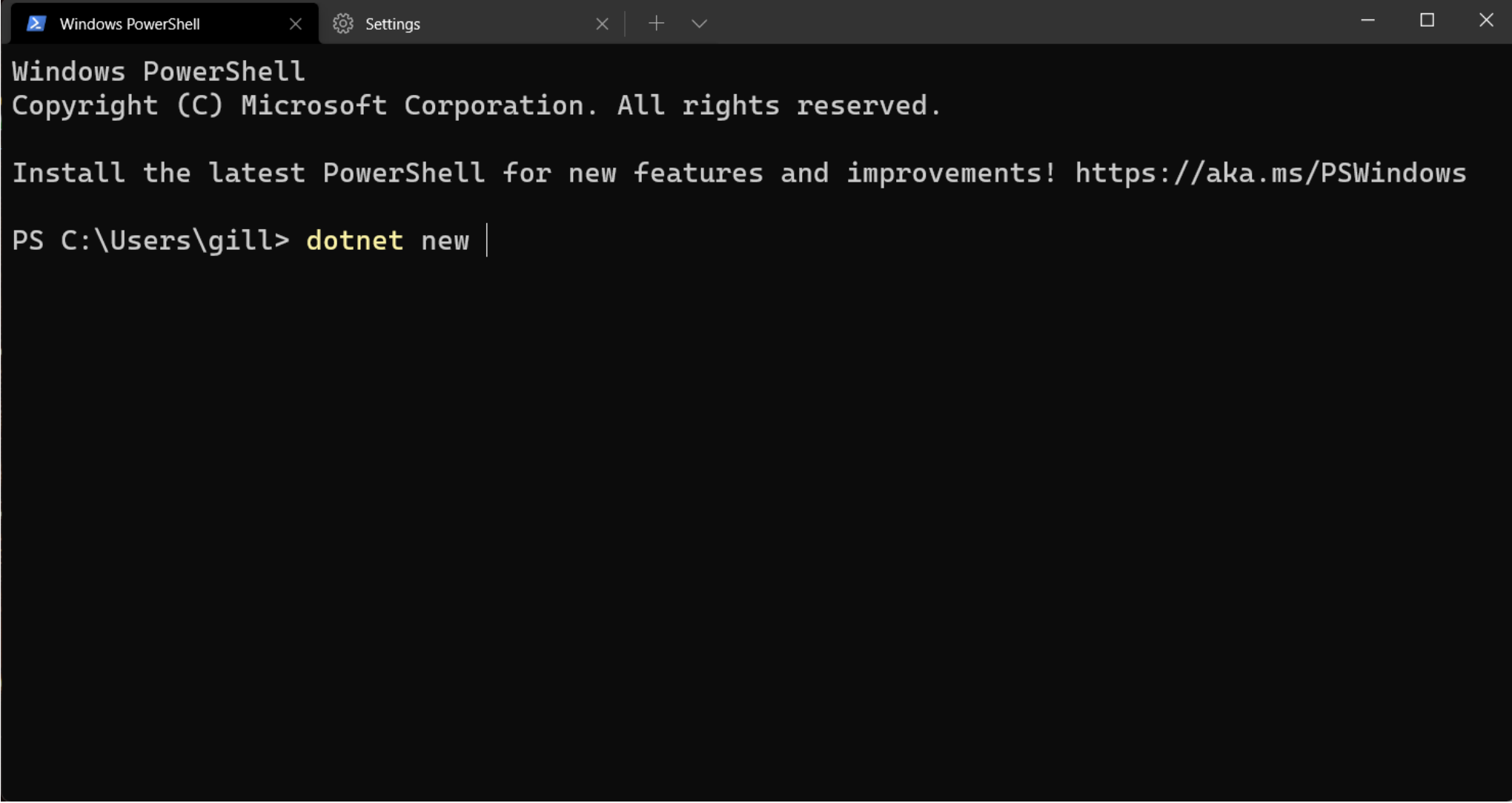
Command-line interface for .NET

Cross-platform tools

“dotnet” command with parameters



Understanding the CLI



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\gill> dotnet new |
```



Creating a New Project

Create a new app

Name of app

`dotnet new console -n "FirstProgram"`

Type of app



```
dotnet new
```

```
dotnet build
```

```
dotnet run
```

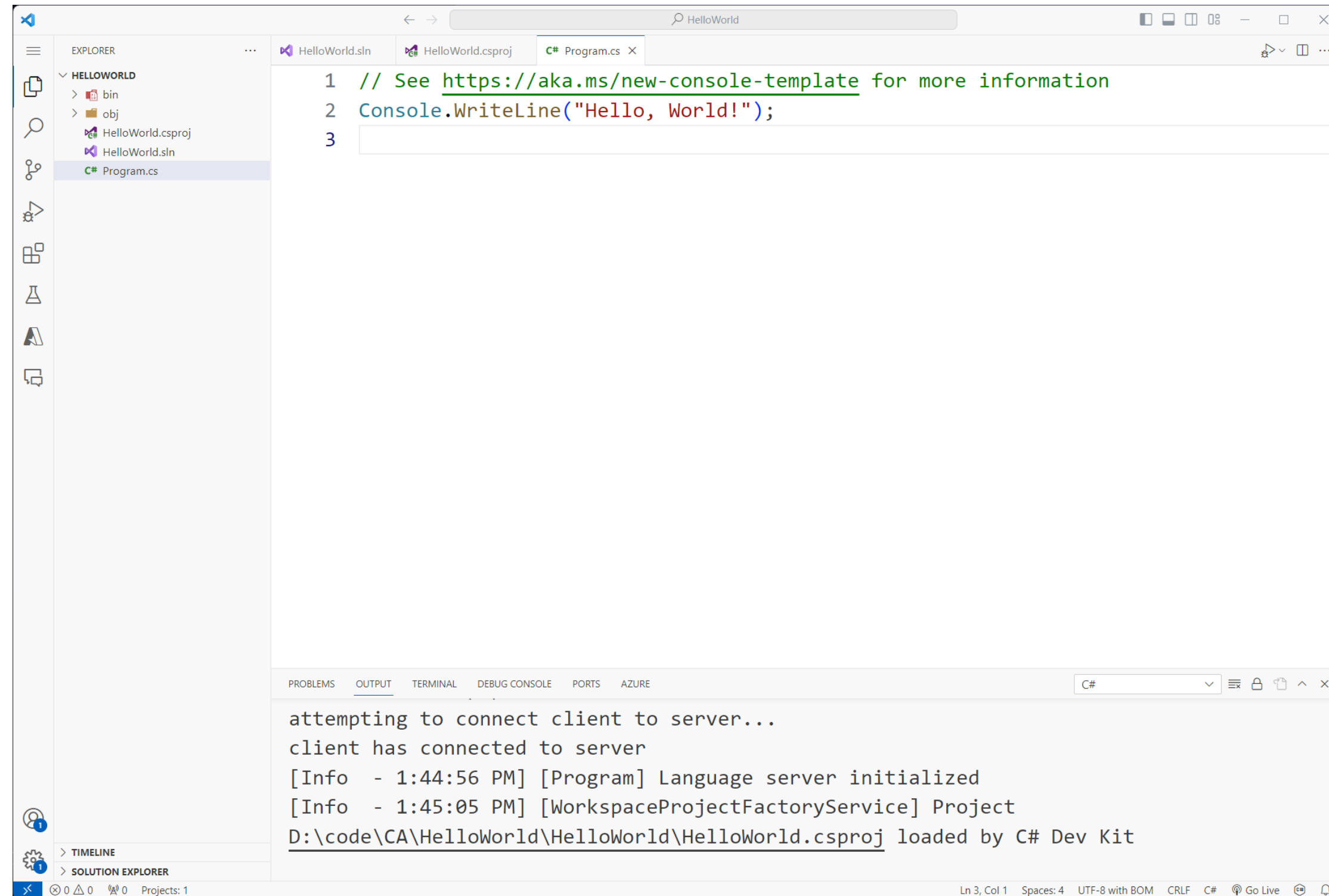
◀ **Create a new project**

◀ **Compile your application**

◀ **Execute your application**



Writing Code with Visual Studio Code



Demo



Installing the required tools

Creating an application using the CLI

Editing code with VS Code

Compiling using the CLI

Running with VS Code and the CLI



Using C# Dev Kit in VS Code

The screenshot displays the Visual Studio Code interface with the Extensions Marketplace open. The 'C# Dev Kit' extension by Microsoft is selected and its details are shown in the main panel. The extension is version 1.2.5, marked as a pre-release, and has 1,708,814 downloads and a 5-star rating from 130 reviews. It is currently installed globally. The left sidebar shows a list of other C# extensions, including 'C#', 'C# XML Documentation', 'C# Extensions', 'C# Snippets', 'Auto-Using for C#', 'C# Namespace', 'C# Tools for Go', 'eppz! (C# them...', and 'C# to TypeScript'. The main panel for 'C# Dev Kit' includes a description, a 'Quick Start' section with three steps, and a 'Feature List & Walkthrough' section with a bulleted list of features. The right sidebar shows 'Categories' (Programming Languages, Debuggers, Testing, Linters), 'Extension Resources' (Marketplace, License, Microsoft), and 'More Info' (Published, Last released, Last updated, Identifier).

EXTENSIONS: MARKETPLACE

C#

C# Dev Kit v1.2.5 Pre-Release
Microsoft | 1,708,814 | ★★★★★ (130)
Official C# extension from Microsoft
Disable Uninstall Switch to Release Version
This extension is enabled globally.

C# Dev Kit for Visual Studio Code

C# Dev Kit helps you manage your code with a solution explorer and test your code with integrated unit test discovery and execution, elevating your C# development experience wherever you like to develop (Windows, macOS, Linux, and even in a Codespace).

This extension builds on top of the great C# language capabilities provided by the [C# extension](#) and enhances your C# environment by adding a set of powerful tools and utilities that integrate natively with VS Code to help C# developers write, debug, and maintain their code faster and with fewer errors. Some of this new tooling includes but is not limited to:

- C# project and solution management via an integrated solution explorer
- Native testing environment to run and debug tests using the Test Explorer
- Roslyn powered language service for best in-class C# language features such as code navigation, refactoring, semantic awareness, and more
- AI-Assisted development

Quick Start

1. Install C# Dev Kit (The [C# extension](#), the [IntelliCode for C# Dev Kit](#), and the [.NET Install Tool](#) will automatically be installed)
2. Open a folder/workspace that contains a C# project (.csproj) and the extension will activate.
3. Check out the [C# Getting Started](#) documentation to learn more!

Feature List & Walkthrough

- Project System / Solution Explorer
 - Solution Node Actions
 - Add Project
 - Build/Run Project
- Code Editing (Uses the [C# extension](#))
 - Refactoring
 - Code Navigation (Go To Definition/References)
 - Code Completions
 - Roslyn powered semantic awareness
- Package Management
 - Automatic NuGet Restore

Categories

Programming Languages Debuggers Testing Linters

Extension Resources

[Marketplace](#)
[License](#)
[Microsoft](#)

More Info

Published	2023-06-06, 20:21:00
Last released	2023-12-14, 20:52:54
Last updated	2023-12-24, 15:35:10
Identifier	ms-dotnettools.csdevkit



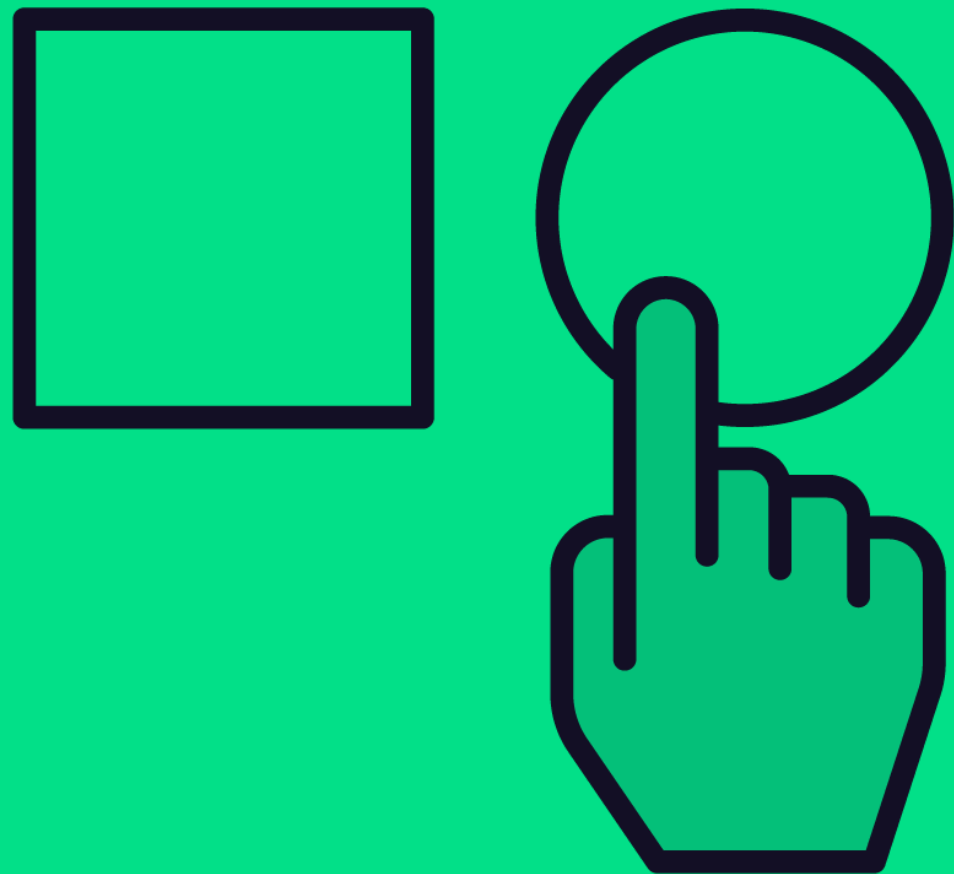
Demo



Using the C# Dev Kit in VS Code

- Creating a new project
- Writing C# code
- Running and debugging our code





The choice is yours!

No matter what platform, you can
write **C#** applications!





Browsing the Docs



Demo



Finding information in the docs



Summary



C# is an object-oriented and type-safe language to write .NET applications

C# is actively maintained

C# can be used to build all types of .NET applications

Applications can be created using

- Visual Studio
- VS Code and CLI



Up Next:

Learning the C# syntax

